

INTRODUCTION TO STOCHASTIC PROCESSES

Siva Athreya, notes by Ramdas Singh

Fifth Semester

Contents

1	RANDOM WALKS	1
1.1	The Setup	1
	Index	3

Chapter 1

RANDOM WALKS

July 22nd.

Quizzes will account for 20% of the final grade, and assignments for 5%. The mid-term will account for 25% of the final grade, and the final exam will account for 50%. The primary reference for random walks is H. Follner.

1.1 The Setup

We wish to concretify the idea of a random walk on the integers $\mathbb{Z}$. Ω is our sample space, with $0 < |\Omega| < \infty$ (we shall later look at the case when Ω is countably infinite). $\mathcal{F}$ is the event space, and is exactly $\mathcal{P}(\Omega)$, the power set of Ω . The intuition of a random walk is as follows; stand at the origin. At each step, flip a fair coin. If the coin comes up heads, take a step to the right; if tails, take a step to the left. Repeat this process N times. S_n here then denotes the position of the random walk after n steps, for $0 \leq n \leq N$.

For a fixed $N \in \mathbb{N}$, we let $\Omega_N = \{\pm 1\}^N$, and $\mathcal{F}_N = \mathcal{P}(\Omega_N)$. For $1 \leq k \leq N$, define the random variable $X_k : \Omega_N \rightarrow \{\pm 1\}$ as $X_k(\omega) = \omega_k$, where $\omega = (\omega_1, \omega_2, \dots, \omega_N) \in \Omega_N$; this models the step taken by the random walk at step k . Further, define $S_0 = 0$, and for $1 \leq n \leq N$, define $S_n : \Omega_N \rightarrow \mathbb{Z}$ as $S_n(\omega) = \sum_{k=1}^n X_k(\omega)$; this models the position of the random walk after n steps. Finally, define the probability measure $\mathbb{P}_N$ on $(\Omega_N, \mathcal{F}_N)$ as $\mathbb{P}_N(A) = \frac{|A|}{2^N}$ for $A \in \mathcal{F}_N$. This models the fair coin flip at each step of the random walk.

Definition 1.1. The sequence of random variables $\{S_n\}_{n=0}^N$ on $(\Omega_N, \mathcal{F}_N, \mathbb{P}_N \equiv \mathbb{P})$, as prescribed above, is called the *simple symmetric random walk* of length N on $\mathbb{Z}$, starting at the origin.

Given $1 \leq k_1 < k_2 < \dots < k_n \leq N$, and $x_i \in \{\pm 1\}$,

$$\mathbb{P}(X_{k_1} = x_1, \dots, X_{k_n} = x_n) = \frac{|\{\omega \in \Omega_N \mid X_{k_i}(\omega) = x_i, 1 \leq i \leq n\}|}{2^N} = \frac{2^{N-n}}{2^N} = 2^{-n}. \quad (1.1)$$

That is, fixing n steps of the walk, the probability of any particular sequence of steps is 2^{-n} . Using this, we can show that for any $1 \leq k \leq N$, $\mathbb{P}(X_k = 1) = \mathbb{P}(X_k = -1) = \frac{1}{2}$. Moreover, we can extract that

$$\mathbb{P}(X_k = x_k, X_l = x_l) = 2^{-2} = \mathbb{P}(X_k = x_k)\mathbb{P}(X_l = x_l), \quad (1.2)$$

showing independence of subsequent steps; the model seems like a reasonable fit for our intuition of a random walk.

Example 1.2. The simple symmetric random walk has independent increments; that is, given $0 < k_1 < k_2 < N$, the random variables $S_k - S_0$, $S_{k_2} - S_{k_1}$, and $S_N - S_{k_2}$ are mutually independent. This is because the increments are functions of disjoint sets of independent random variables X_k , and hence are independent themselves. Similarly, one may concretely show that for $1 \leq k < m \leq N$, the distribution of $S_m - S_k$ is the same as that of S_{m-k} .

The following properties of the simple symmetric random walk are immediate consequences of the above definition and example.

Suppose we have $\mathbb{P}(S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \dots, S_0 = 0) > 0$. Then, we are free to compute the conditional probability $\mathbb{P}(S_n = a_n \mid S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \dots, S_0 = 0)$. One may verify that

$$\mathbb{P}(S_n = a_n \mid S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \dots, S_0 = 0) = \mathbb{P}(S_n = a_n \mid S_{n-1} = a_{n-1}), \quad (1.3)$$

for all $a_1, \dots, a_n \in \mathbb{Z}$ such that $\mathbb{P}(S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \dots, S_0 = 0) > 0$. This is called the *Markov property* of the simple symmetric random walk. In words, this property states that given the present state of the walk, the future evolution of the walk is independent of its past history.

Now suppose $\mathbb{P}(S_k = a) > 0$ for some $k \geq 1, a \in \mathbb{Z}$, and let $m > k, b \in \mathbb{Z}$. Then, one can simply compute that $\mathbb{P}(S_m = b \mid S_k = a) = \mathbb{P}(S_{m-k} = b - a)$.

The next question we may ask is, on an average, where the random walk is located after n steps, for $1 \leq n \leq N$. We can compute the expected value of S_n as follows:

$$\mathbb{E}[S_n] = \mathbb{E}\left[\sum_{k=1}^n X_k + 0\right] = \sum_{k=1}^n \mathbb{E}[X_k] = \sum_{k=1}^n 0 = 0. \quad (1.4)$$

Thus, to get a better idea of the location of the random walk after n steps, we compute the variance of S_n :

$$\text{Var}(S_n) = \text{Var}\left(\sum_{k=1}^n X_k + 0\right) = \sum_{k=1}^n \text{Var}(X_k) = \sum_{k=1}^n 1 = n. \quad (1.5)$$

So, via the central limit theorem, we can approximate for large n as

$$\mathbb{P}\left(-3 \leq \frac{S_n}{\sqrt{n}} \leq 3\right) \approx \Phi(3) - \Phi(-3) \quad (1.6)$$

where Φ is the cumulative distribution function of the standard normal distribution.

We now claim that

$$\mathbb{P}(S_{2n} = 0) = \binom{2n}{n} 2^{-2n} = \mathbb{P}(S_{2n-1} = 1) = \mathbb{P}(S_{2n-1} = -1). \quad (1.7)$$

To this end, we note that for $S_n = x$, we must have taken $\frac{n+x}{2}$ steps to the right, and $\frac{n-x}{2}$ steps to the left. Thus, we have

$$\mathbb{P}(S_n = x) = \frac{|\{\omega \in \Omega_N \mid S_n = x\}|}{2^N} = \binom{n}{\frac{n+x}{2}} \frac{2^{N-n}}{2^N} = \binom{n}{\frac{n+x}{2}} 2^{-n}. \quad (1.8)$$

The binomial coefficient attains its maximum at $x = 0$ for even n , and at $x = \pm 1$ for odd n . Thus, we have shown the above required claim. We can further approximate this via *Stirling's formula*, which states $k! \sim \sqrt{2\pi k} k^k e^{-k}$ for large k . For now, we shall treat this as an equality. Thus, for large n ,

$$\mathbb{P}(S_{2n} = 0) = \binom{2n}{n} 2^{-2n} \sim \frac{\sqrt{4\pi n} (2n)^{2n} e^{-2n}}{2\pi n n^{2n} e^{-2n}} 2^{-2n} = \frac{1}{\sqrt{\pi n}}. \quad (1.9)$$

We now show that given any interval $[a, b] \subset \mathbb{R}$, the probability that the random walk stays in this interval is zero as $n \rightarrow \infty$. To this end, note that either $\mathbb{P}(S_n = x) \leq \mathbb{P}(S_n = 0)$ or $\mathbb{P}(S_n = x) \leq \mathbb{P}(S_n = 1)$ holds for any given n ; moreover, both of the quantities on the right-hand side are bounded by $\frac{c}{\sqrt{n}}$ for an appropriate constant $c > 0$. Thus, we have

$$0 \leq \mathbb{P}(a \leq S_n \leq b) = \sum_{x=a}^b \mathbb{P}(S_n = x) \leq \frac{c(b-a+1)}{\sqrt{n}} \rightarrow 0 \text{ as } n \rightarrow \infty. \quad (1.10)$$

Index

Markov property, 2

simple symmetric random walk, 1

Stirling's formula, 2